

Who Validates the Validators? A Measurement-Science Audit of Benchmark-Validity Diagnostics

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Abstract

We present ValidEval, an offline-safe framework for validating benchmark-validity diagnostics before using them to support benchmark claims. The framework pairs controlled flaw injection, diagnostic evidence gates, and claims ledgers with real-data stress tests. In a HELM MMLU / MMLU-Redux stress test, broad matrix-derived diagnostics did not strongly recover documented issue labels, illustrating why diagnostic-specific validation and blocked-claim reporting are necessary. On a 39-model HELM MMLU panel, we find substantial subject-level ranking sensitivity while proxy diagnostic-weighted rankings remain close to accuracy rankings. Direct/hash MMLU-Redux alignment, decoupled synthetic validation, second-benchmark evidence, and full parametric 2PL remain unresolved gates rather than paper claims.

1 Introduction

Validity is not accuracy. A benchmark score can be high while the benchmark measures shortcuts, contamination, scoring artifacts, saturation, or low-discrimination items rather than the claimed construct.

ValidEval treats benchmark-validity diagnostics as measurement instruments. Diagnostics should support benchmark claims only after they have evidence about sensitivity, false-positive behavior, specificity, uncertainty, and materiality under a documented protocol.

The current empirical spine is a public HELM MMLU response matrix with 39 models and 14,042 scored items. This panel supports artifact-backed analyses of panel validity, proxy psychometric item diagnostics, subject-level ranking sensitivity, and resampling baselines. The paper treats these analyses as measurement evidence about diagnostics and rankings, not as a certificate that MMLU is valid or invalid.

The MMLU-Redux case remains a weak/negative external-validation stress test. Direct or hash-backed alignment is not confirmed for the current files, and broad matrix-derived diagnostics did not strongly recover structurally aligned MMLU-Redux labels. The claims ledger therefore blocks detection-success claims.

The contributions are:

1. A measurement-validity framework for benchmark diagnostics with explicit evidence states: supported, weak, blocked, or not run.
2. Artifact-backed real-panel analyses of the active 39-model HELM MMLU matrix, including panel validity, proxy IRT diagnostics, subject-level rank sensitivity, and baselines.
3. A claims-ledger and evidence-gating workflow that prevents unsupported validity claims.

4. A MMLU-Redux stress test showing that broad proxy diagnostics do not automatically validate against structurally aligned external issue labels.
5. An offline-safe toolkit and reproducible audit artifact format for benchmark authors and evaluators.

2 Related Work

ValidEval is not another leaderboard runner; it is a diagnostic-validation and claim-gating framework for benchmark-validity diagnostics.

The framework builds on measurement-validity and psychometric traditions [Messick, 1995, American Educational Research Association et al., 2014], but applies them to cached language-model benchmark artifacts rather than treating a benchmark score as self-interpreting. It complements evaluation runners such as HELM and lm-evaluation-harness [Liang et al., 2022, Gao et al., 2023]: those systems make model evaluation reproducible, while ValidEval asks what benchmark-validity claims the resulting response matrix can support.

The current artifact spine uses MMLU and MMLU-Redux as a stress test [Hendrycks et al., 2021, Gema et al., 2024]. GSM8K, BBH, TruthfulQA, GPQA, and MMLU-Pro are treated as candidate or deferred benchmark families, not as completed evidence in this draft [Cobbe et al., 2021, Suzgun et al., 2022, Lin et al., 2022, Rein et al., 2023, Wang et al., 2024].

3 Validity Framework

Validity is treated as a multidimensional evidence profile rather than a scalar score, following the measurement principle that score interpretations require evidence for the intended use rather than only high reliability or high accuracy [Messick, 1995, American Educational Research Association et al., 2014]. A diagnostic can support a paper claim only when the artifact trail specifies the construct, the response panel, the scoring rule, the diagnostic family, uncertainty or baseline checks where available, and the claim boundary.

The framework separates three states that are often conflated. First, a benchmark can have a valid response matrix for a diagnostic without proving the benchmark measures its intended construct. Second, a diagnostic can flag suspicious items without proving those items are externally confirmed errors. Third, a ranking can be sensitive to subjects or item subsets without becoming a true model ranking. ValidEval records these distinctions in claims ledgers and evidence-state tables.

The active paper evidence uses this gate discipline: the 39-model MMLU panel passes the local panel-validity gate, proxy IRT diagnostics are available, subject-level rank sensitivity is artifact-backed, and MMLU-Redux direct/hash validation remains blocked.

4 The valideval Toolkit

The toolkit is built around cached response matrices and reproducible artifact directories. For the MMLU case study, ValidEval imports wide prediction records, builds a model-by-item correctness matrix, runs a panel-validity gate, computes proxy psychometric item diagnostics, and writes ranking, subject-instability, diagnostic-disagreement, and baseline artifacts.

The release workflow also includes claim ledgers, blocked-claim reports, paper tables and figures, Kaggle execution packaging, and reviewer-packet tooling. Scaffolds are kept separate from evidence:

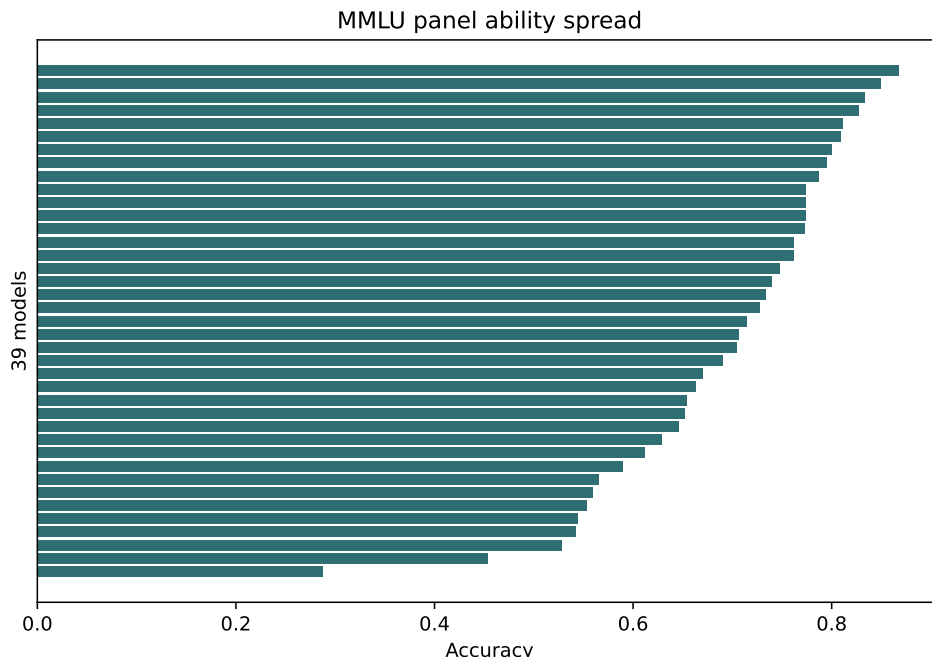


Figure 1: Model accuracy spread across the active 39-model HELM MMLU panel.

decoupled synthetic validation, second-benchmark runs, full 2PL, and direct/hash MMLU-Redux alignment remain marked as `RESULT_REQUIRED` or blocked unless a saved artifact proves that the run occurred.

5 Audit Protocol

The audit protocol starts by freezing the active evidence state. For this draft, the active MMLU panel is `cache/mmlu/wide/matrix.csv`, derived from the HELM wide prediction details. Legacy three-model MMLU files are retained only as provenance. All analyses are run against cached local artifacts; no paid API calls or downloads are required for the reported MMLU results.

The protocol then applies gates in order: panel shape and missingness, panel-validity, proxy item diagnostics, ranking and subject-instability analyses, baseline comparisons, and external validation attempts. Missing or blocked steps are reported explicitly. In this run, direct/hash MMLU-Redux alignment is blocked, decoupled synthetic validation is blocked by missing preregistration guard artifacts, and second-benchmark execution is deferred to the Kaggle package.

6 Results

Active panel. The reconciled active MMLU panel is the 39-model HELM wide matrix with 14,042 scored items and zero missingness. The panel-validity gate passes with ability spread 0.580. Historical three-model files are provenance only, not the active panel.

Proxy psychometrics. Proxy IRT diagnostics were computed for all 14,042 items. The proxy run flags 1,037 negative-discrimination items, 1,342 near-zero-discrimination items, and 2,676 extreme-

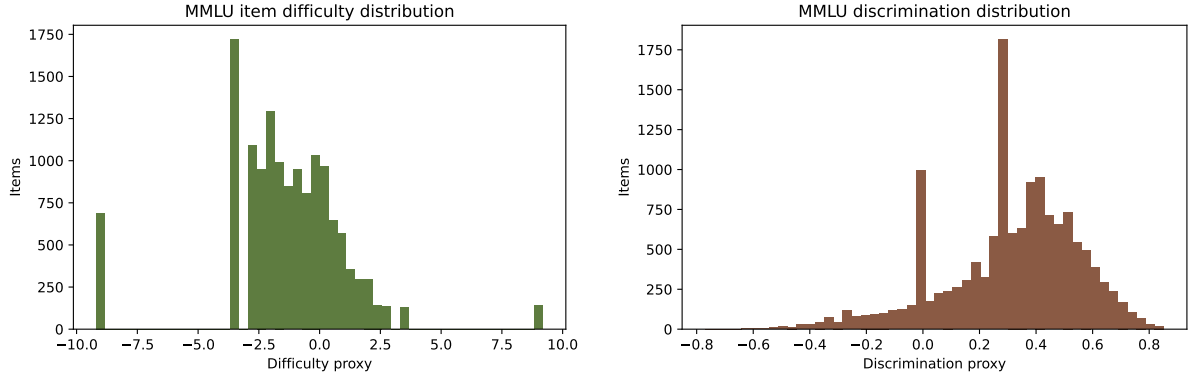


Figure 2: Proxy item difficulty and discrimination distributions on the active MMLU panel.

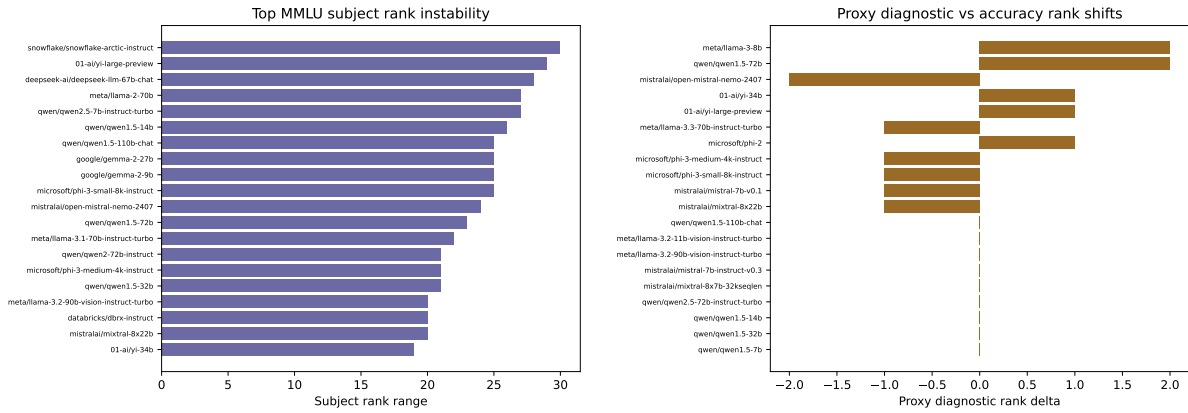


Figure 3: Subject-level rank instability is larger than proxy diagnostic-vs-accuracy rank shifts in this run.

difficulty items. Rasch/1PL is not available for this wide matrix under the local BFGS implementation, and the 2PL run exports proxy slopes only.

Ranking sensitivity. Subject-level ranks vary substantially: every model has subject rank range at least 3, and the maximum subject rank range is 30. In contrast, proxy diagnostic-weighted ranking is very close to accuracy ranking, with Spearman correlation 0.998 and maximum absolute rank delta 2.

Baselines. Random-item and subject-stratified bootstrap baselines are stable at the top of the ranking. Across 1,000 iterations per baseline, top-1 match rate is 1.0 and the 95th percentile max absolute rank delta is 3.0.

MMLU-Redux. Direct/hash alignment remains blocked: the current files have zero direct item-id matches and no stable metadata or hash match path. The available structural alignment maps 370 labels by `subject_numeric_index`. Existing structural metrics remain weak/negative (AUROC 0.539, AUPRC 0.0289, precision@10 0.0), so detection-success claims are blocked.

Analysis	Spearman	Max rank delta	Claim state
Proxy diagnostic vs. accuracy	0.998	2	proxy only
Subject instability	–	30	subject sensitivity

Table 1: Ranking sensitivity summary. Proxy diagnostic weighting is close to accuracy ranking; subject ranks vary more.

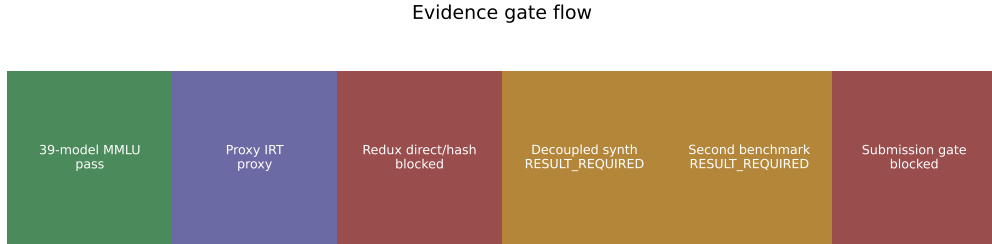


Figure 4: Current evidence gates. Green and purple states are artifact-backed; blocked and result-required states remain claim blockers.

Deferred evidence. Decoupled synthetic validation, second-benchmark evidence, Kaggle output import, full Rasch/1PL, and full parametric 2PL remain **RESULT_REQUIRED**.

7 Analysis

The evidence is consistent with a measurement-science caution: benchmark validity diagnostics need their own validation and baselines before they can license benchmark claims. The active MMLU panel is large enough for the implemented proxy diagnostics, but the external MMLU-Redux stress test remains weak/negative and structurally aligned rather than direct/hash confirmed.

The ranking result is also deliberately bounded. Subject-level rank sensitivity is large enough to be paper-relevant, but proxy diagnostic-weighted rankings stay close to accuracy rankings. This means the current evidence supports a claim about subject sensitivity under one public MMLU panel, not a claim about the true ordering of models.

The blocked gates are part of the result. Decoupled synthetic validation was not run because guard artifacts are missing. Second-benchmark validation was not run because no local wide panel exists. The Kaggle notebook creates an execution path, but evidence remains **RESULT_REQUIRED** until outputs are imported and validated.

8 Limitations

MMLU-Redux alignment is structural, using `subject_numeric_index` with confidence 0.85, not direct-id or hash-confirmed item matching. The MMLU-Redux result is weak/negative external validation: broad/proxy diagnostics did not strongly recover the structurally aligned labels, and subject-normalized evidence weakened the earlier raw label-error signal.

The MMLU IRT output is proxy-only and should not be described as full psychometric IRT or full 2PL. The local Rasch/1PL optimizer is blocked for the active wide matrix because the

implementation would fit 14,080 free parameters; the 2PL output is a proxy-slope export only. The narrow raw label-error signal is at most a preliminary review-queue cue and does not support detection-success claims.

Synthetic validation can be circular when generators mirror detectors. The paper should therefore report the mixed cross-flaw specificity and held-out generator results alongside false-positive/error-control checks and materiality labels. Synthetic-harness power, numeric calibration with confidence/logprob outputs, and human-reviewed real-benchmark threshold calibration remain [RESULT REQUIRED].

Real benchmark findings remain preliminary and protocol-scoped. The second-benchmark path is prepared through a Kaggle notebook package, but no Kaggle outputs were imported in this run. Domain packs are not paper-validated unless separately supported. GPQA local-panel findings are protocol evidence, not broad validity evidence.

The subject-instability finding may depend on the model panel, HELM extraction, subject taxonomy, and MMLU item mix. It should be treated as a reproducible panel finding, not a universal property of all benchmarks or model rankings.

9 Ethics and Broader Impact

Validity diagnostics can be misused if their outputs are treated as scalar certificates or as automatic proof that a benchmark is defective. ValidEval is designed to reduce that risk by pairing every artifact with a claim state and by preserving blocked results.

The release excludes raw benchmark text from paper-facing artifacts and reviewer packets. Reports use ids, aggregate metrics, and sanitized tables. This protects dataset provenance while still allowing reviewers to audit which artifacts support each claim.

The broader goal is not to discard benchmarks based on one diagnostic, but to make benchmark claims more precise: what construct is being measured, which threats have evidence, which diagnostics are validated, and which claims remain unresolved.

10 Conclusion

ValidEval provides audit evidence under documented protocols, not a certificate of benchmark truth. The current artifact set supports a cautious real-panel finding on the 39-model HELM MMLU matrix: subject-level rankings are sensitive to subject composition, while proxy diagnostic-weighted rankings remain close to accuracy rankings.

The same artifact set also shows why claim gates matter. MMLU-Redux direct/hash alignment is blocked, structural Redux validation remains weak/negative, decoupled synthetic validation and second-benchmark evidence are still **RESULT REQUIRED**, and the project is not yet a NeurIPS Datasets & Benchmarks ready package.

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